

From Demand Forecasts To Replenishment Decisions

Demand Forecasting & Inventory Analytics

UE Digital Analytics im Handel, JKU Linz · Session T2, 03.10.2026 · Sina Mirshahi

About Me

- I am **Sina Mirshahi**, a researcher at Logistikum, University of Applied Sciences Upper Austria, in Steyr
- I work on applied AI and predictive analytics with industrial and logistics partners: forecasting, inventory, computer vision, and AI assistants that answer from company documents
- I am a Ph.D. candidate in Engineering Informatics at Tomas Bata University, where I taught neural networks and data science
- Recent papers: Forecasting Without Fine-Tuning, ready-trained forecasting models against traditional methods (Procedia Computer Science 277, 2026); Two-Step Framework for Intermittent Demand Forecasting (Procedia Computer Science 278, 2026)

Questions are always welcome, here or by email: Sina.Mirshahi@fh-steyr.at

Where This Afternoon Fits

Session	Lecturer	The question	What you leave with
T1, 03.10. morning	Patrick	What are real life use cases for retail analytics, and what are the data requirements?	Understanding of the importance of analytics in retail and the relevance of the clean data
T2, 03.10. afternoon	Sina	How do we forecast demand and derive the order quantity?	A forecast choice and a service target, each with one number and one limitation
T3, 04.12.	Taha	Which orders may come back, and what should we do?	Return-risk prediction and the cost of prediction errors

- This afternoon has four chapters: understand the demand, forecast it, evaluate the forecast, turn it into stock. Two hands-on blocks in Colab, one break

→ **Today you turn a retail forecast into an order and a service target you can defend.**

What You Will Take Home

Rather than coding from scratch, we look at how a decision becomes code: a rule the computer runs and you can read and question.

- **Recognise** what kind of demand a product has, and know what error to expect and which methods to try before anyone builds a model
- **Know the main methods**, from repeating last year to smoothing, methods for products that sell rarely, and one model for the whole assortment, and what each can and cannot see
- **See sales as model inputs**: how weekly sales become the numbers a model learns from, and how a model's settings are tuned
- **Read an accuracy number** and know what it hides: why the most common one fails on zeros, why a useless forecast can score well, why the direction of the error matters as much as its size
- **Ask how a forecast was tested**: how many test dates, how far ahead, with which metric, and whether the future was allowed in
- **Understand a model's reasons** for one forecast, as factors you can check against your own experience
- **Turn a forecast into stock**: how large a cushion to hold, how much to order, and what a higher service target buys

→ **Reading forecasts and acting on them, with the mechanics understood, not only the vocabulary.**

Google Colab And Gemini

A notebook in your browser: text, code and results on one page, run on Google's computers. Free with a Google account.

- **A notebook** is a page of cells: text cells explain, code cells run and show their result right below
 - **Colab** runs the code on a Google computer in the cloud: nothing to install, and your copy is saved in your Google Drive
 - **Run** one cell with the play button next to it, or Shift + Enter; **Runtime, Run all** runs the whole notebook from the top
 - **Gemini**, Google's AI assistant, is built in if your account offers it: ask in plain English to explain an output or an error, or to write code from a description
- Check Gemini against the numbers: it can sound sure and still be wrong. Give it only the course data, never company or personal data

Other notebook tools	Where it runs	Good for
Jupyter Notebook, JupyterLab	your own computer; free, open source	the original; Colab uses its file format, .ipynb
Kaggle Notebooks	the browser; free, also from Google	public datasets and competitions
VS Code with Jupyter	your own computer; free	notebooks next to regular programs
Databricks, Microsoft Fabric	company cloud platforms	notebooks on a company's own data

The same .ipynb notebook file opens in all of them.

Try it now, before the course notebook

Open colab.research.google.com and sign in, choose **New notebook**, type 2 + 2 in the code cell and press **Shift + Enter**. Then open **Gemini** and ask: "What does a code cell do?"

- **A notebook keeps the words, the code and the results on one page; Gemini helps you read it, and you check Gemini.**

Loading Data Into Colab

Three ways to get a file into Colab. Each ends in one line of code that reads the file into a table.

Route	What you do	The code	Keep in mind
1. Upload a file	Files panel (the folder icon on the left), upload, pick the file	<code>pd.read_csv("sample_weekly_sales.csv")</code>	The file is deleted when the session ends
2. From a direct link	Paste the address of the file itself; on GitHub, the Raw button	<code>pd.read_csv("https://raw.githubusercontent.com/.../sample_weekly_sales.csv")</code>	A Drive share link is not a direct link. Our notebooks load their data this way
3. From Google Drive	Put the file in a Drive folder, then connect Drive (Colab asks for permission)	<code>drive.mount("/content/drive")</code> <code>pd.read_csv("/content/drive/MyDrive/JKU_T2/sample_weekly_sales.csv")</code>	The file stays in your Drive between sessions

- **Then explore:** `df.head()` shows the first rows, `df.describe()` the numbers at a glance. Or ask Gemini to summarise and plot the table, and check what it did

→ Upload only data you are allowed to share: files in Colab and in your Drive sit on Google's servers

Try all three with the sample file (three products, weekly store sales): [notebook 0 in Colab](#).

→ **A file on your computer: upload it. A file on the web: use its link. A file you want to keep: put it in Drive.**

Open Your First Notebook Now

The notebooks and the data live at github.com/FH-Prevail/jku_course. You need a Google account. No account, or it does not work? Work with a partner.

- **Open:** go to the repository (link on the board, or scan the code) and click the **Open in Colab** button next to notebook 1 (**1 · Demand Patterns and Forecast Comparison**) in the table on that page
 - Nothing happens? Open the notebooks folder, click **1_see_and_forecast.ipynb** and use the button at its top, or in Colab: **File, Open notebook, GitHub**, type FH-Prevail/jku_course
- **Make it yours:** click **Copy to Drive** (or File, Save a copy in Drive). Your copy keeps whatever you change
- **Run it:** Runtime, **Run all**. The first cell, **Start here**, fetches the course data; after a few seconds it prints **Ready** right under itself
- **Read:** the tables and pictures are the lesson. Use the dropdowns and sliders; nothing else needs editing
- **Ask:** Gemini explains any output ("explain this table in plain English"); check it against the numbers
- Accidentally changed a cell? Edit, Undo, then run it again. Restarting does not undo edits; a fresh copy from the repository does



The repository: the course folder on GitHub

What you should see under the first cell (Start here) after Run all

Ready. The course data is loaded.
The forecasts were computed in advance, using only what was known at each forecast date.

→ **Everyone should see the word Ready before we go on.**

Understand The Demand

Channels, seasons, promotions,
empty shelves, and four kinds of
demand

01

Meet Our Retailer

Artificial data, built like a real retailer's: one row per product, channel and week.

Three rows of data/classroom/demand.csv. The first two: one product, one week, two channels, so two series. The third: a planned promotion, 15 % off.

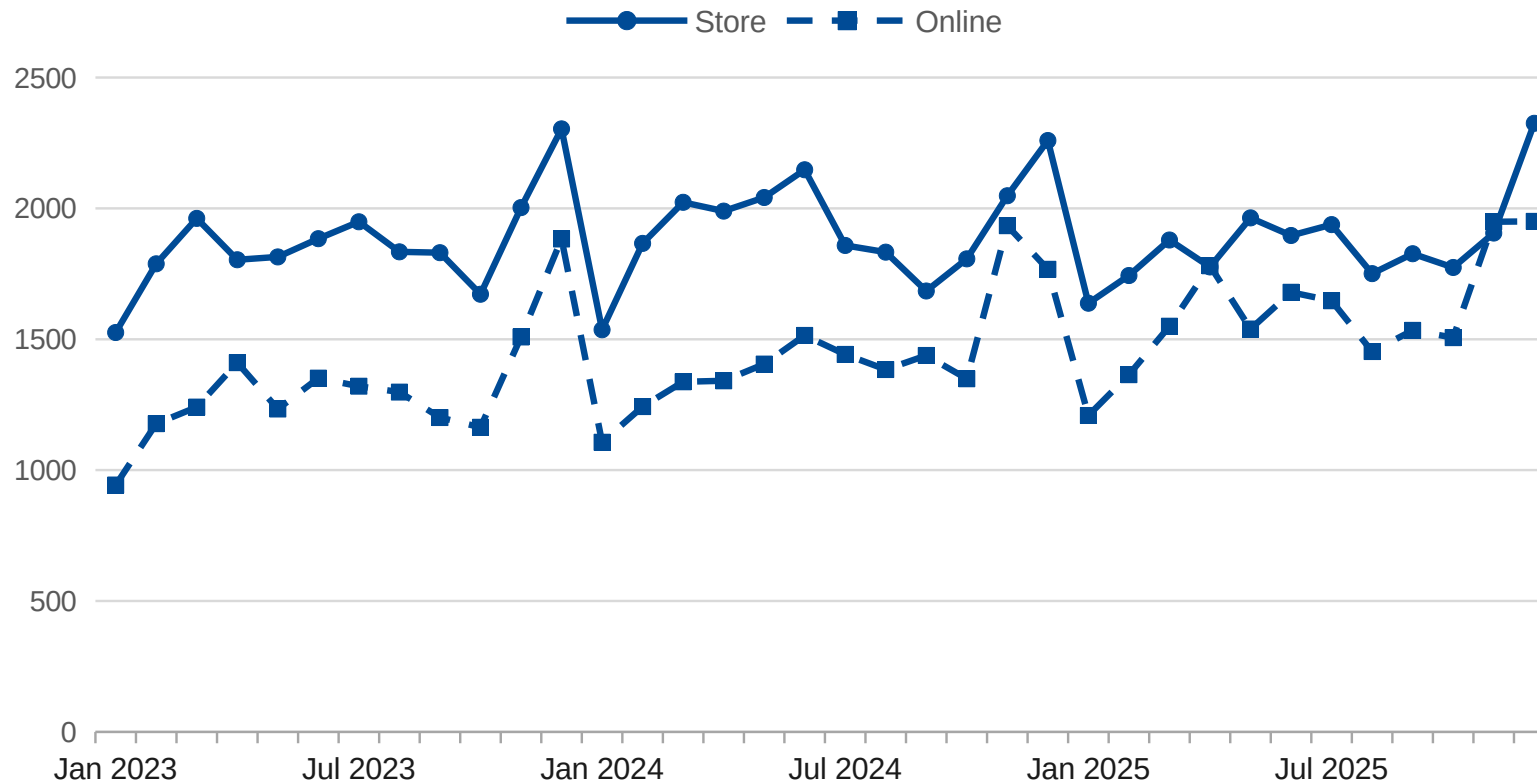
week_start	product_name	channel	units	price	discount_pct	promo_flag	stockout_flag
2023-01-02	Plush Bear	store	77	25.00	0	0	0
2023-01-02	Plush Bear	online	57	25.00	0	0	0
2023-04-24	Plush Bear	store	177	21.25	15	1	0

What we have	What it means
40 products in 2 channels	80 series: each product in store and online
156 weeks, 518,571 units	2023-01-02 to 2025-12-22, weeks starting Monday
Plush Bear, Puzzle 1000, Vacuum Filter	Our three recurring examples: regular, variable and sparse sales
1,037 promotion weeks with their discount	Planned in advance: information we have before we order
4 flagged stock-out weeks	Sales were zero because the shelf was empty; demand was not zero

→ A product in a channel is one series of weekly sales. We forecast series.

Sales By Channel

All 40 products together, average units sold per week in each month, 2023 to 2025.



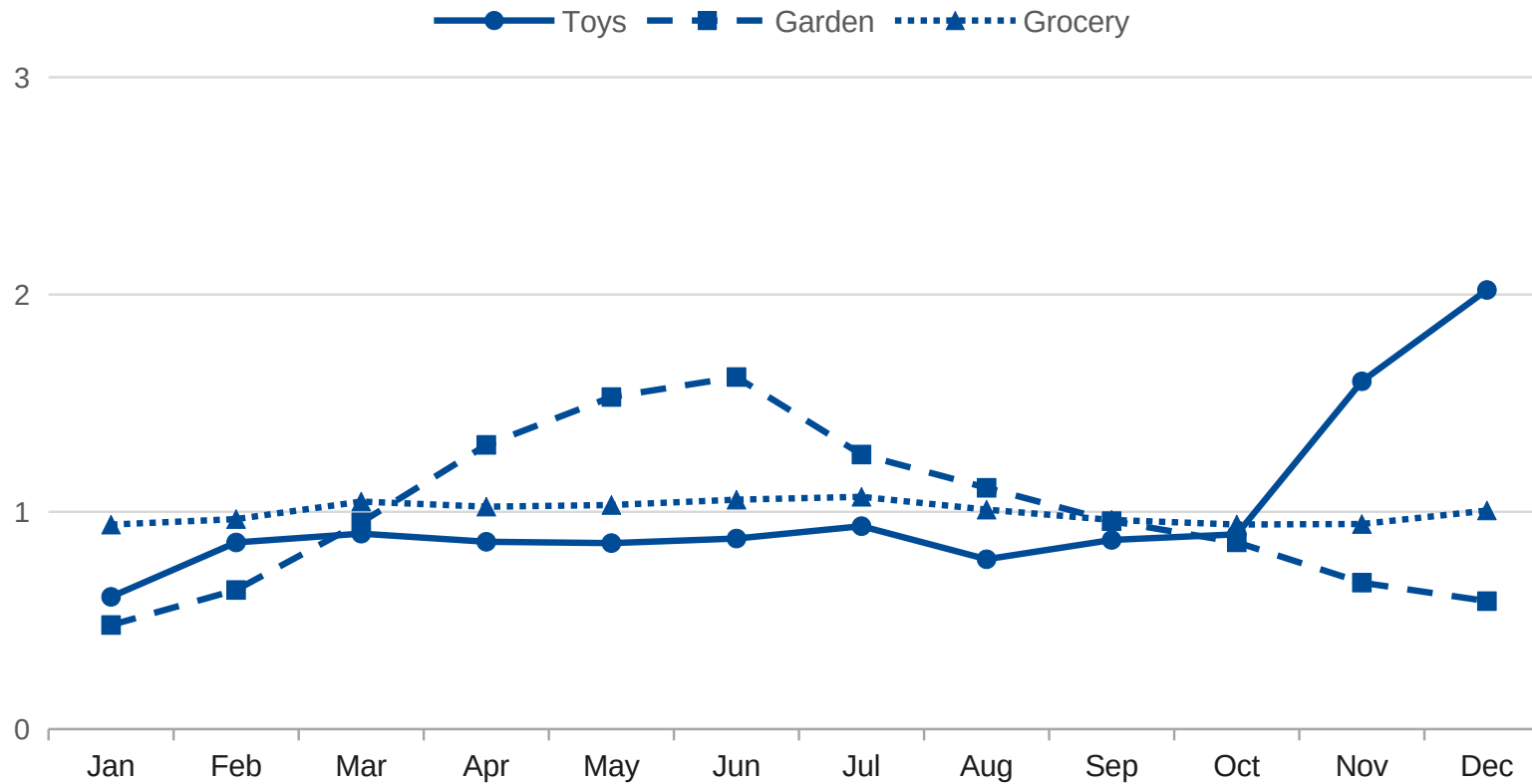
Year	Store	Online
2023	1,855	1,300
2024	1,919	1,437
2025	1,868	1,596

Average units per week in each year. Online grows every year; the store moves up and down. A single product can look different from its channel total.

→ Do not assume that both channels follow the same trend.

The Busy Season By Category

Seasonal index: a month's average weekly sales divided by the category's average weekly sales. 1.00 is the category's average month.

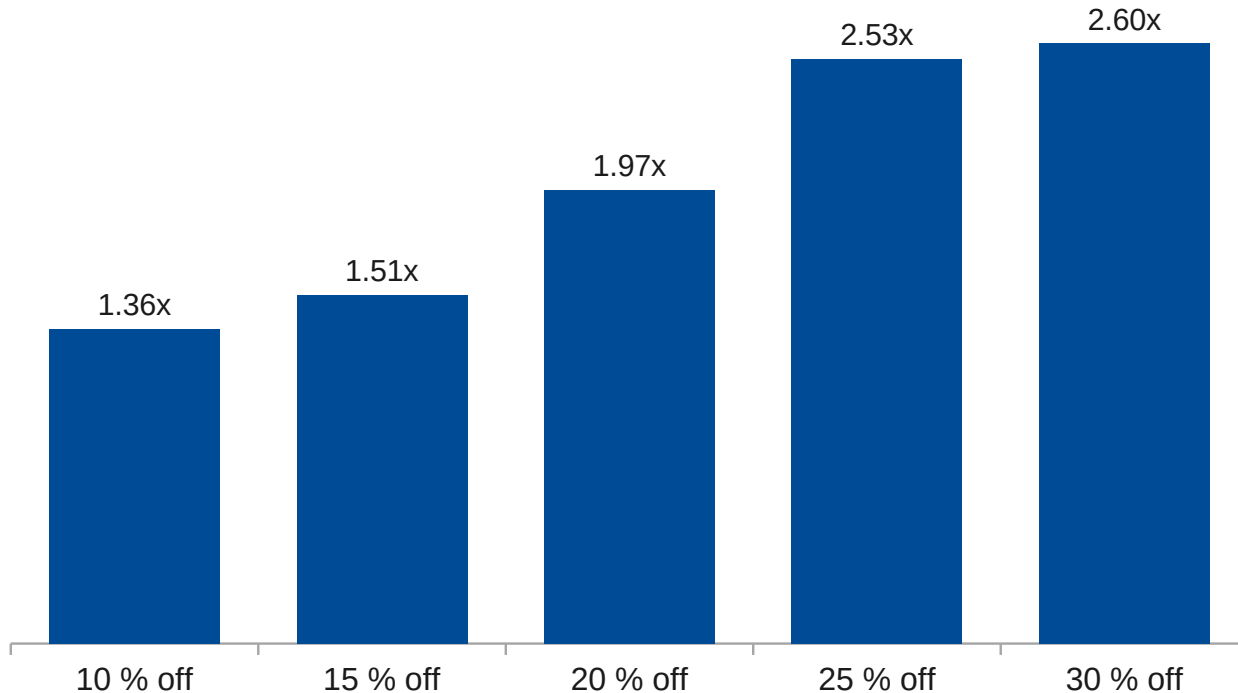


- **Toys** sell 2.0x their average month in December (Electronics, not shown: 1.5x)
- **Garden** peaks in June at 1.6x and drops to 0.5x in January
- **Grocery** stays between 0.94 and 1.07: almost no season
- A method that cannot see the calendar will be surprised every December

→ When the pattern repeats every year, the forecast should expect it.

Promotion Effects By Discount

Sales in a promotion week compared with a normal week of the same product, by discount. 1.00x is a normal week.



Normal week: no promotion this week or the week before.

→ **Planned promotions belong in the information the forecast is given.**

- A deeper discount sells more: from **1.36x** at 10 percent to **2.60x** at 30 percent
- Online reacts more than the store: **2.10x** against **1.91x**
- The week after a promotion sells **9 percent below normal**: a sign that some customers bought earlier, during the promotion
- Promotions are **planned**, so a forecast can know about them in advance. How much of the extra sales is new demand, and how much only moved forward, needs a separate test

Zero Sales And Empty Shelves

What we observe	What we know	What to do
No sale, stock on the shelf	Demand was zero	Keep the week as it is
No sale, shelf empty	Demand is unknown	Flag the week; estimate it from the weeks around it, or leave it out of the scoring
Future stock availability	Nobody knows it in advance	Never use a stock-out flag as an input for future weeks

- In our data **4 weeks** are flagged. When the models learn from the history (training), each one is replaced by the average of the four weeks before it; when we measure forecast errors (scoring), they are left out
- Real point-of-sale data often has no flag. Several zero weeks in a row for a product that usually sells deserve a closer look

→ Zero sales can mean no demand, or no stock. Tell the two apart before you forecast.

Four Demand Patterns

Two numbers per series: ADI (average demand interval), the average weeks between sales; CV^2 (squared coefficient of variation), how irregular the sale sizes are.

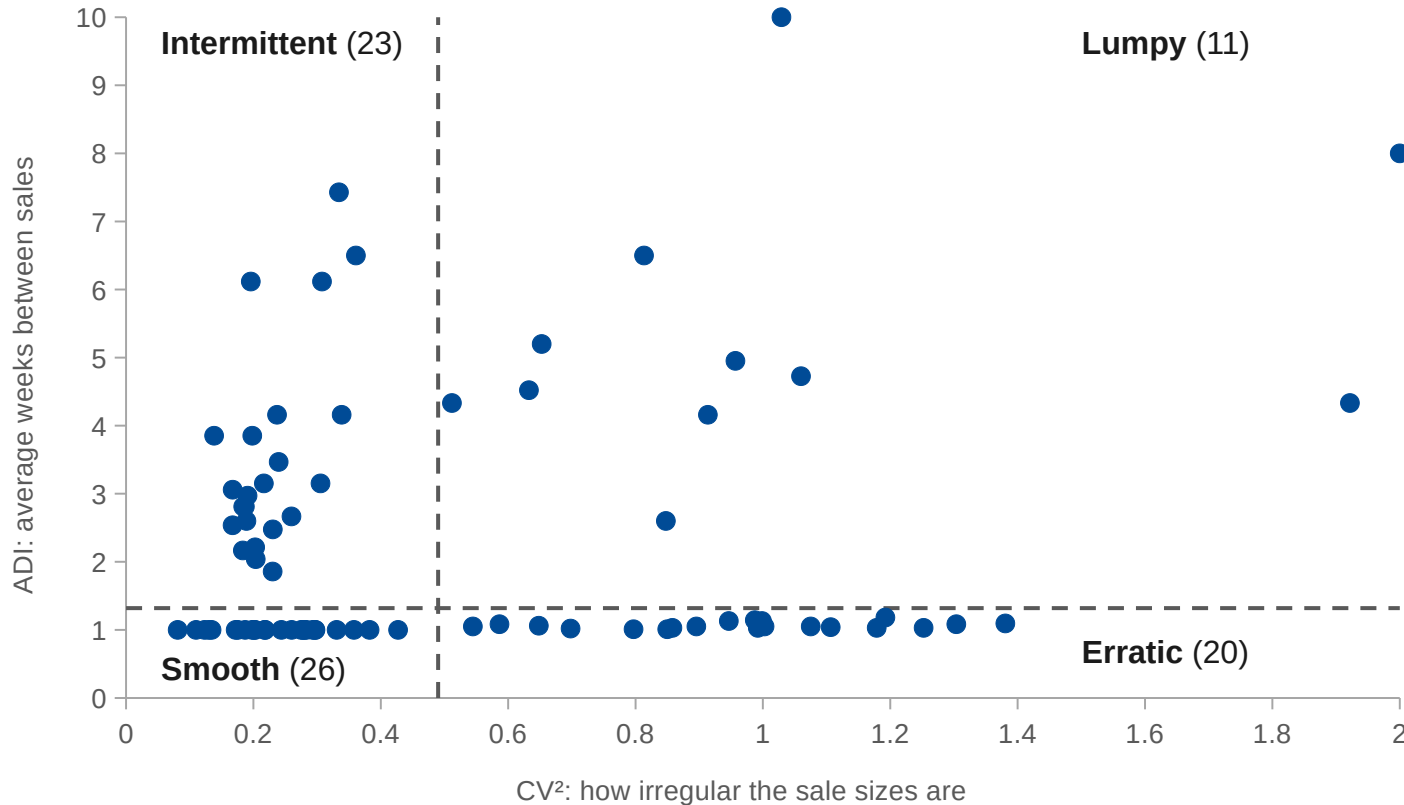
	CV ² below 0.49: regular sizes	CV ² at least 0.49: irregular sizes
ADI at least 1.32: many weeks without a sale	INTERMITTENT: 23 series, 0.4 % of the units	LUMPY: 11 series, 0.3 % of the units
ADI below 1.32: a sale most weeks	SMOOTH: 26 series, 89 % of the units	ERRATIC: 20 series, 11 % of the units

- The limits 1.32 and 0.49 are a convention from the literature (Syntetos and Boylan), not a law. You still test which method works
- 34 of 80 series are slow movers: intermittent or lumpy. In 2025 they sold under one percent of the units. The classes use only 2023 and 2024, the history a forecast would have had

→ The pattern tells you what error to expect and which method to try first; it does not choose the method for you.

Four Patterns, One Picture

Every series placed by its two numbers, measured on 2023 and 2024. The dashed lines are the conventional limits: CV^2 0.49 and ADI 1.32.



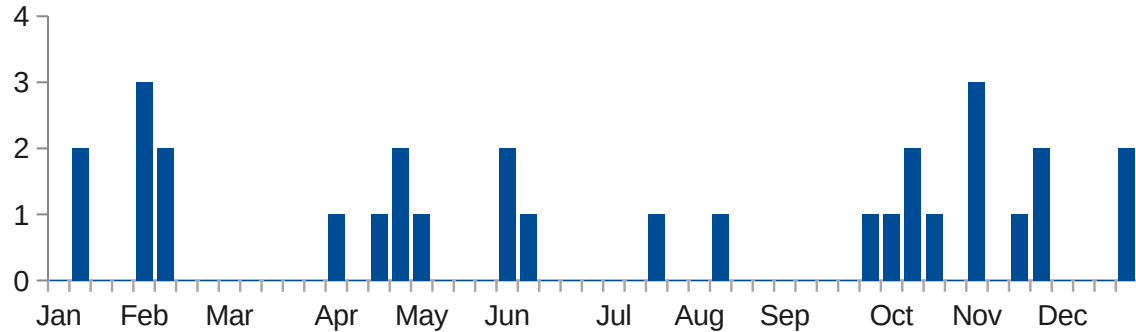
- **Each dot is one product in one channel** (80 dots; the two values outside the picture are drawn on its edge)
- **Further right**: the amounts sold vary more from one sale to the next
- **Further up**: more weeks pass between sales. ADI 1 means a sale every week; ADI 4 means a sale every four weeks on average
- The fast movers crowd the bottom edge; the slow movers sit above the dashed line, most between ADI 2.5 and 5

→ The higher a dot sits, the less a weekly forecast can tell you.

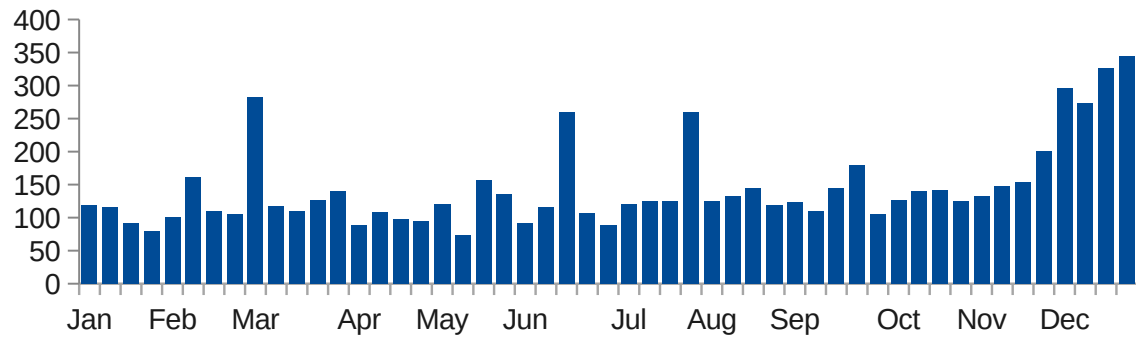
Pattern Examples

Weekly units in 2025 for one product of each kind, placed in the same corners as on the picture.

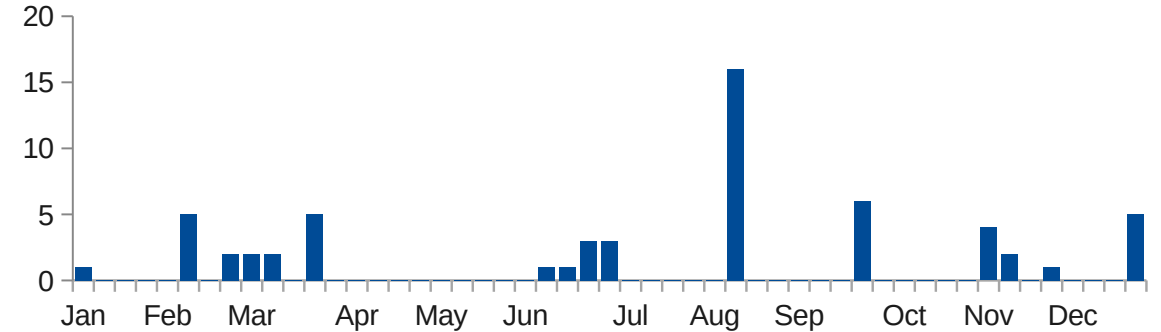
Intermittent: Vacuum Filter, store. ADI 2.54, CV^2 0.17 (2023 and 2024); in 2025 63 % zero weeks, 0.6 units a week



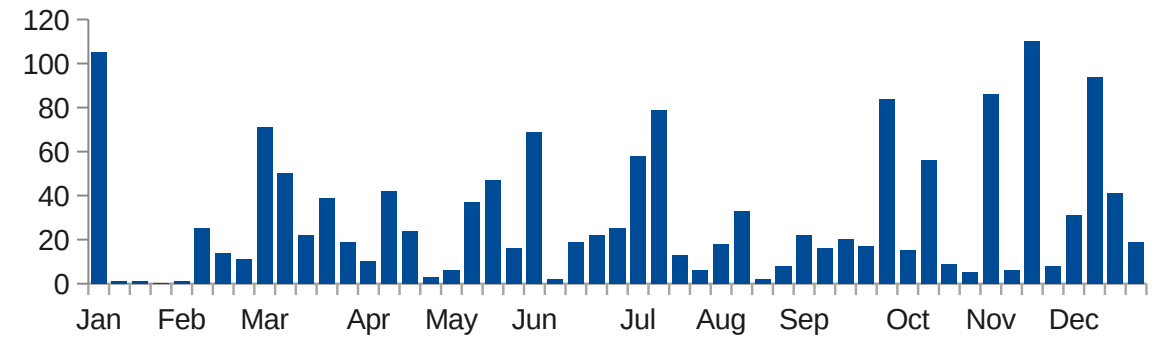
Smooth: Plush Bear, store. ADI 1.00, CV^2 0.30 (2023 and 2024); in 2025 0 % zero weeks, 145.4 units a week



Lumpy: Trimmer Blade, store. ADI 2.60, CV^2 0.85 (2023 and 2024); in 2025 69 % zero weeks, 1.2 units a week



Erratic: Puzzle 1000, store. ADI 1.03, CV^2 1.18 (2023 and 2024); in 2025 2 % zero weeks, 30.1 units a week



Forecast

Three approaches, what they use,
and how they compare on the
same four weeks

02

Three Approaches Worth Comparing

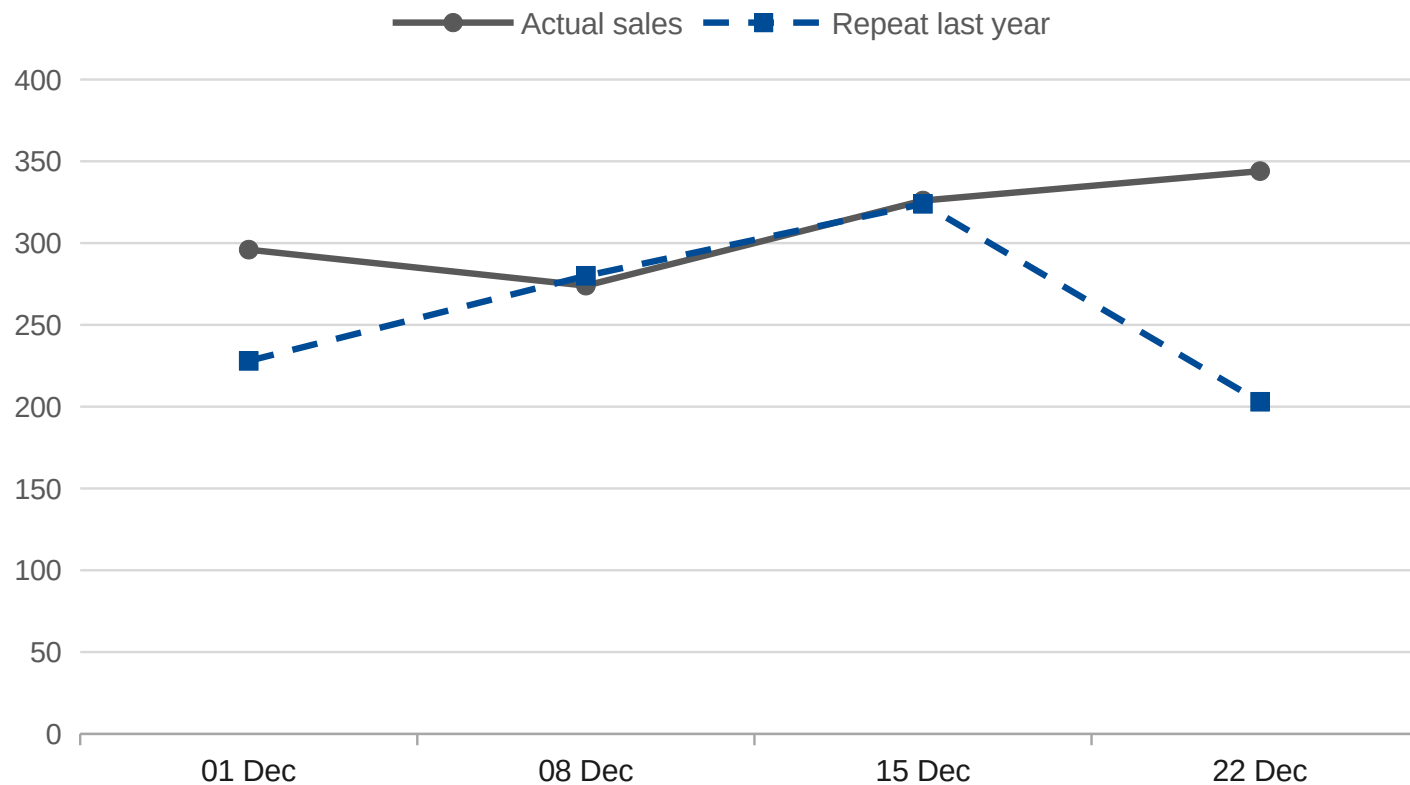
Approach	What it uses	Main limitation
Repeat last year (seasonal naive)	The same week 52 weeks earlier	Repeats last year's unusual weeks too, and cannot see a changed promotion
Smooth recent demand (simple exponential smoothing, SES)	A weighted average that trusts the newest weeks most; the forecast is a flat level	This version does not anticipate a season
Global model (LightGBM: many small decision trees; one model for all 80 series)	Shared history of all products, plus calendar, product identity, planned price and promotion	More preparation, harder to explain, and a win is not guaranteed

- The first two are **local** methods: each series is forecast from its own past. The third is **global**: one model learns from all series at once
- The next slides add Holt-Winters and the methods for slow movers. In chapter 3 a replay of 2025, the backtest, scores all of them, plus the average of the last 8 weeks (a moving average). The worked examples and the notebooks use only these three

→ **Keep a simple benchmark in every comparison.**

The Seasonal Naive Benchmark

Plush Bear, store: the four weeks from 1 December 2025, forecast with the sales up to the week of 24 November.



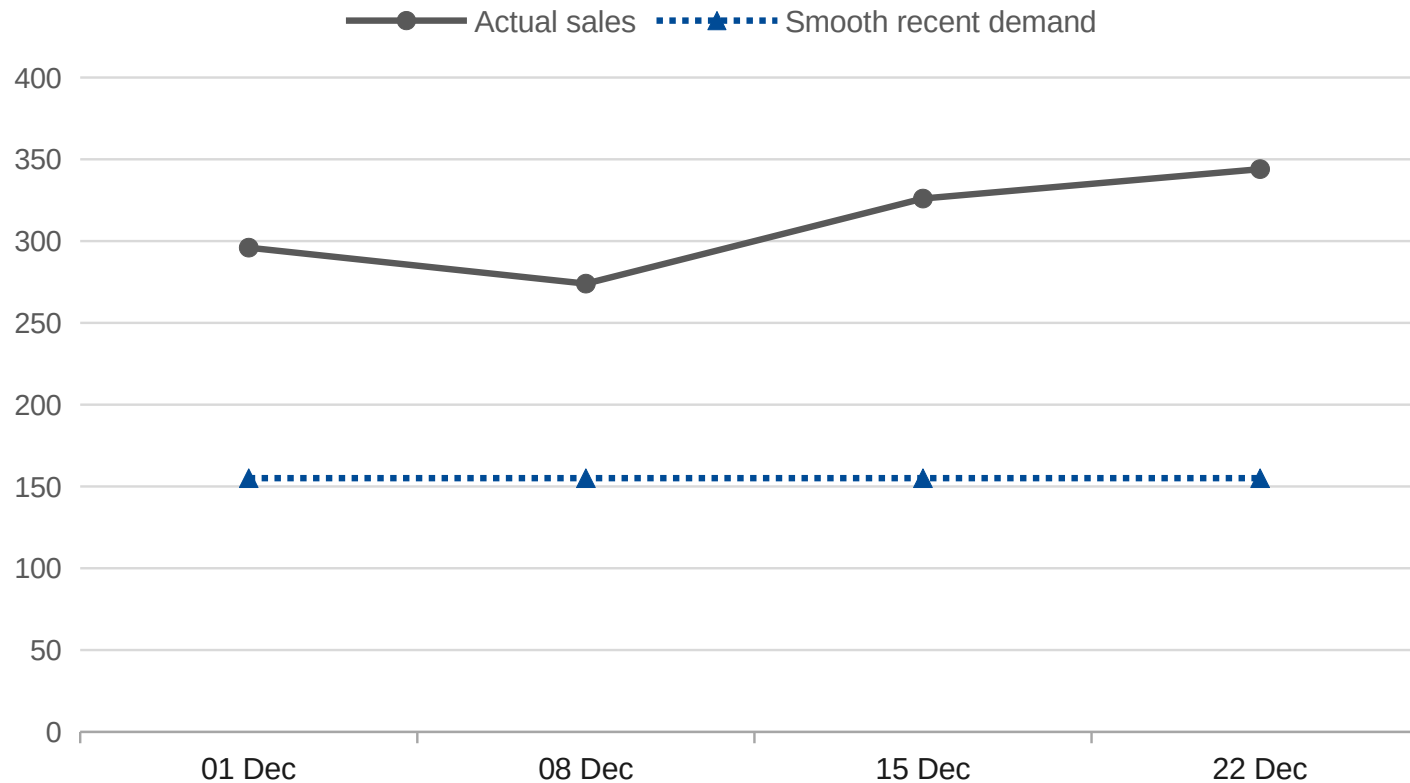
Week	Last year	Actual
01 Dec	228	296
08 Dec	280	274
15 Dec	324	326
22 Dec	203	344

The forecast copies December 2024, week by week. The middle two weeks are almost exact (280 against 274, 324 against 326); the first and the last are far too low. The week of 22 December had a promotion in 2025 and none in 2024: the rule copies last year and cannot see this year's plan.

→ Seasonal naive gives us a clear benchmark to beat.

Smoothing Recent Demand

Same product, same forecast date, same four weeks: the smoothed level of the recent weeks.



A flat 155 units a week. The smoothed level of the past weeks, carried forward. For the bear it follows a changed level within a few weeks, which the seasonal naive cannot do, but it does not know that December sells more.

→ Simple exponential smoothing follows the level, not the coming season.

Exponential Smoothing In Detail

Weighted averages that trust the newest weeks most; Holt-Winters adds a trend and a season.

Method	What it tracks	Update rule	Forecast shape	WAPE, smooth series
SES (simple exponential smoothing)	the level	new level = α x this week's sales + (1 - α) x old level	flat	29.4 %
Holt	level and trend	the same rule again for the week-to-week change	a straight line; damped: flattens out	not in our backtest
Holt-Winters	level, trend, yearly season	a third rule for each week of the year; needs two full years	last year's shape around a moving level	32.4 %

- α (alpha) is the weight of the newest week, between 0 and 1, fitted from each product's history: 0.24 for the Plush Bear, but close to 0 for 65 of our 80 series, whose forecast is then almost the plain average of all past weeks
- **WAPE**, the total miss as a percentage of total demand (lower is better; chapter 3), on the 26 smooth series over 2025: Holt-Winters 32.4 %, SES 29.4 %, seasonal naive 36.6 %. A season learned from one product's two years is a noisy season

→ Adding components does not automatically help: here the flat SES beats Holt-Winters four weeks ahead.

Slow Movers

Smoothing a series that is zero most weeks gives a forecast that jumps after every sale. Croston fixed that in 1972.

Method	Idea	When a product stops selling	WAPE, intermittent	Bias
Croston	Estimate the size of a sale and the gap between sales separately. Forecast = size ÷ gap	Keeps the old rate forever	129.9 %	-4 %
Croston SBA (Syntetos and Boylan)	Croston with a small downward correction, because in theory Croston runs slightly high	Same as Croston	128.4 %	-8 %
TSB (Teunter, Syntetos, Babai)	Update the chance of a sale every week, the size only when a sale happens	Lowers the rate week by week	132.3 %	-4 %

- **Bias** (forecast minus actual, as a share of actual; minus means too low): Croston -4 %, TSB -4 %, SBA -8 %: on our data the correction goes the wrong way. The global model: -15 %
- **WAPE** on the same 23 intermittent series: SES 129 %, Croston, SBA and TSB 128 to 132 %, the global model 121 %. Above 100 percent is normal for a product that sells one unit some weeks

→ The forecast is a weekly rate: add it up over the weeks until an order arrives (the lead time), and judge it by the stock decision.

Local And Global Models

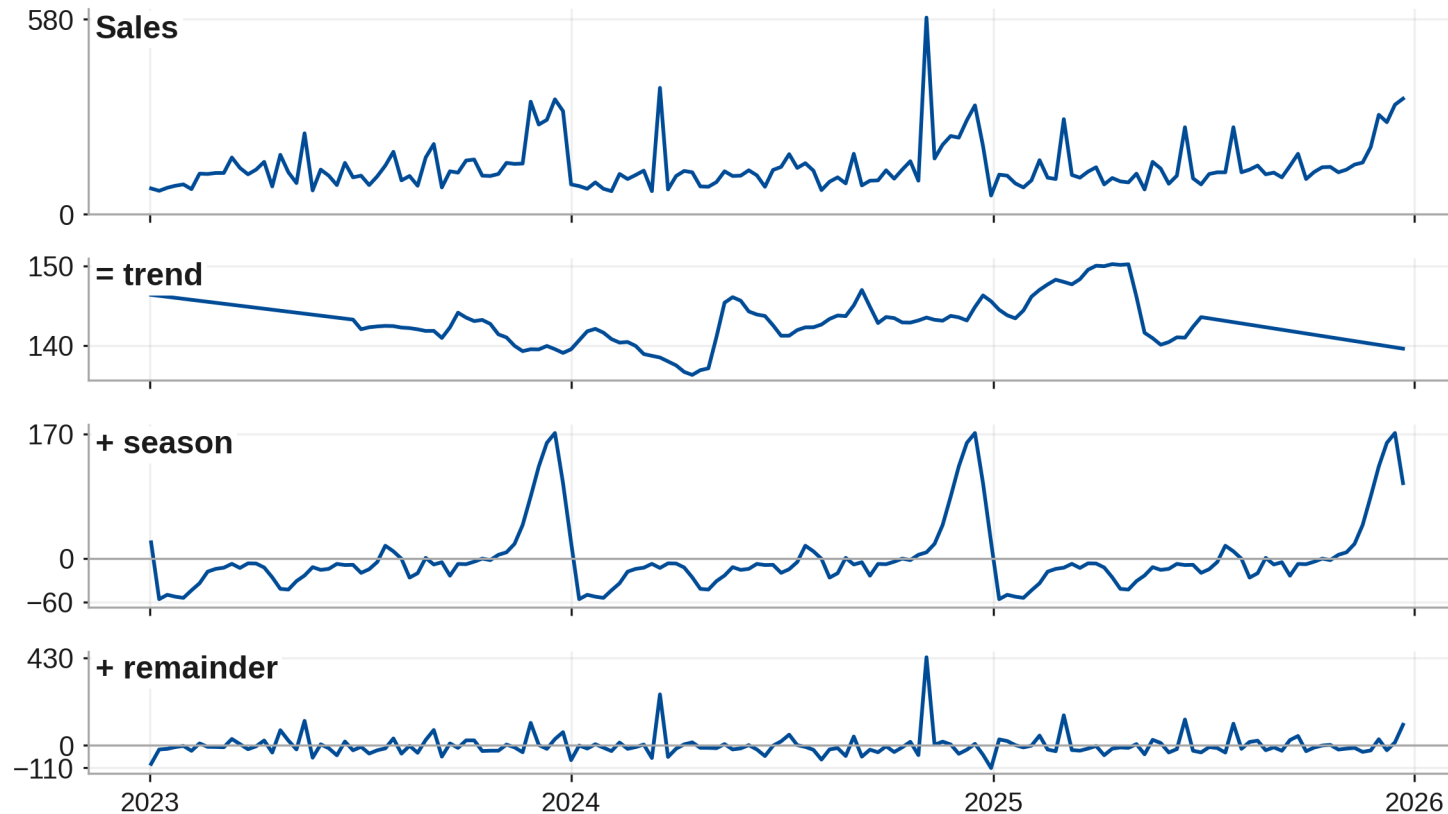
	Local model: one per series	Global model: one for all series
What is fitted	Each series on its own history, about 2 to 3 years of weeks	One model on every row of every series at once
What it can learn	Only what its own series shows: level, trend, season	Patterns that repeat across products: a promotion appears 1,037 times in the data, about 13 times in one series
New or short series	Weak: a new product has no history	Can borrow the season and the promotion effect from similar products (ours still needs a year of history)
Planned inputs	Possible, but each product learns the effect from its own few promotions	Natural: one input column, one effect learned from all products
Cost and risk	Fast, simple, easy to explain per series	More preparation; the big products can dominate it; it needs a backtest before you trust it

- Ours is **LightGBM**, a model built from many small decision trees. It learns rules such as "a 20 percent discount: expect about 2 times a normal week"

→ **Local or global is about the data a model learns from, not about the algorithm.**

A Series In Three Parts

Plush Bear, store, 2023 to 2025: a trend (the 52-week average), a yearly season (the typical deviation of each week) and a remainder.

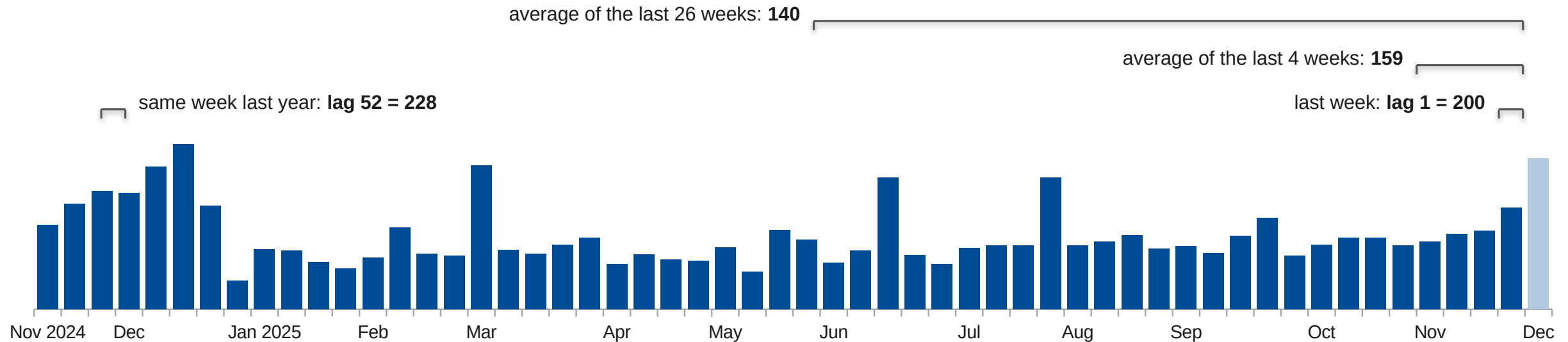


- **Trend:** the level the series sits at and where it is heading, here between 136 and 150 units a week
 - **Season:** the pattern that repeats every 52 weeks. The week before Christmas (week 51) adds 172 units
 - **Remainder:** what trend and season do not explain. In the 13 promotion weeks it averages +120 units, otherwise -12
- A statistician reads a series as parts. A machine-learning model reads it as features: a row of numbers for every week

→ The same weeks can be read as parts that add up, or as a table of features.

From A Series To Features

Plush Bear, store: the forecast made after the week of 24 November, the last week with known sales, for the week of 1 December 2025. The brackets mark where each number comes from.



Light bar: the week we forecast. Its actual sales, 296 units, are the target the model learns to predict; they are unknown when the forecast is made.

- **From the calendar:** week 49 of the year, the Christmas window. **From the plan:** promotion no, price EUR 25.00. **Identity:** Plush Bear, store, Toys

→ A feature is one number the model is given for a week; the model learns how they combine.

One Row Of Features

All 26 features of one row: Plush Bear, week of 1 December 2025, one week ahead, for the global model trained on 2024. Weeks back count from the week being forecast.

From the series itself	Value
sales 1 week back	200
sales 2 weeks back	154
sales 3 weeks back	148
sales 4 weeks back	133
sales 8 weeks back	126
sales 12 weeks back	110
sales 26 weeks back	91
sales 52 weeks back (same week last year)	228
average of the last 4 weeks	159
average of the last 8 weeks	146
average of the last 13 weeks	141
average of the last 26 weeks	140
variation over the last 8 weeks	24.1

More history, then calendar, plan and identity	Value
share of zero weeks, last 13	0 %
weeks since the last sale	0
week of the year	49
month	12
Christmas window (weeks 48 to 51)	yes
Black Friday week (week 47 in our data)	no
promotion planned	no
discount planned	0 %
price planned	EUR 25.00
which product	P010 (Plush Bear)
which channel	store
which category	Toys
weeks ahead	1

→ The model was trained on 28,000 rows like this one: all 80 series, the weeks of 2024, each for 1 to 7 weeks ahead.

Internal And External Features

Internal features come from the company's own data, external ones from outside it. Everything must be known on the day of the forecast.

Where it comes from	Typical features	Known in advance?	In our model
Internal: the product's own history	last weeks' sales, averages over 4 to 26 weeks, weeks since the last sale, share of zero weeks	Yes, by construction	Yes, 15 features
External: the calendar	week of the year, month, public and school holidays, Black Friday, paydays	Yes	Week, month, Black Friday, Christmas
Internal: planned commercial actions	promotions, discount depth, price, in-store display, advertising, a new listing	Yes, if it is planned	Promotion, discount, price
Internal: product and place	category, price band, pack size, lifecycle stage, store size, region, channel	Yes	Product, category, channel
External: outside the company	weather, local events, competitor promotions, economic indicators	Only as forecasts themselves	No
Internal: the stock situation	stock-outs, shelf availability	Not for future weeks	No

Plus one more feature: how many weeks ahead the forecast is (1 to 7). Together the 26 features of slide 26.

→ **Only information available when the forecast is made may be used.**

Choosing The Features

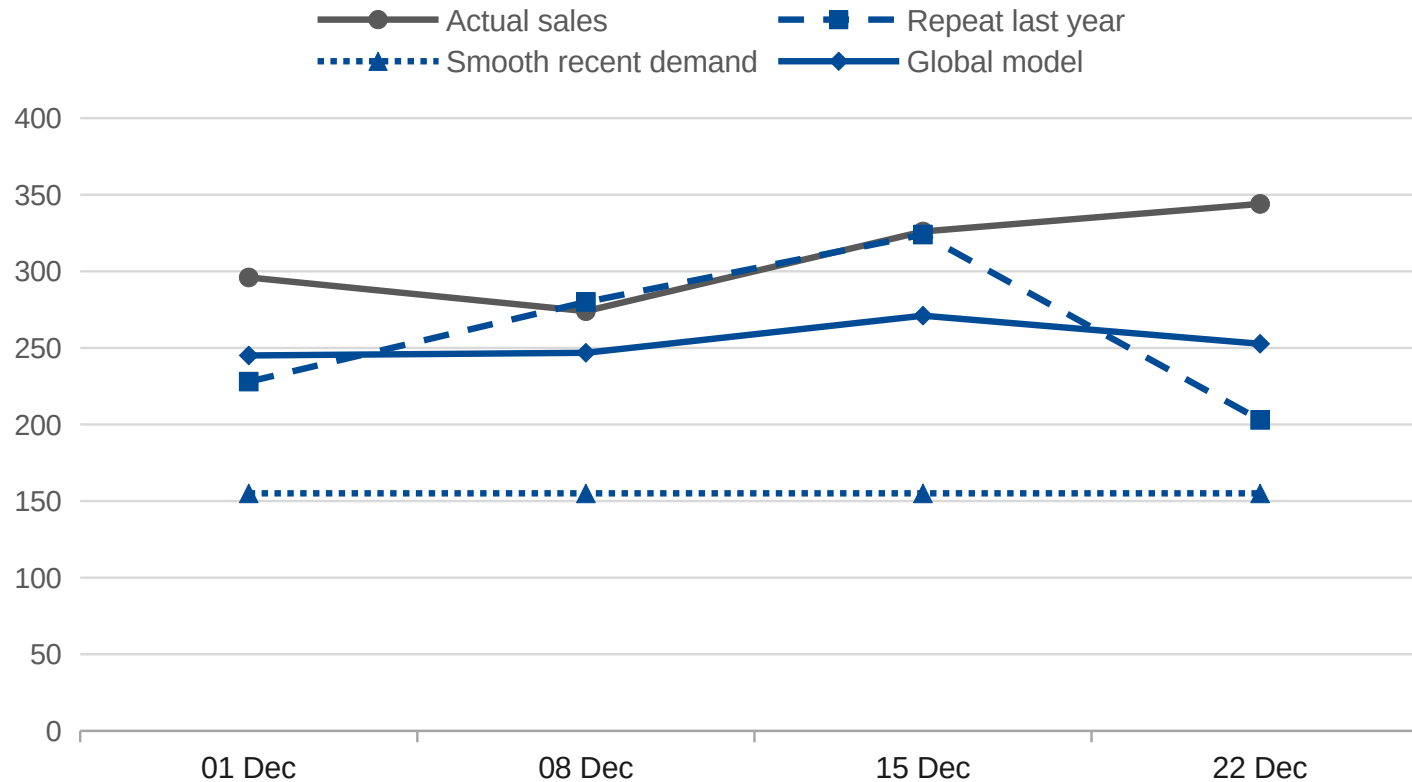
Feature selection: more features is not better. Each one has to earn its place.

- **Leakage.** A feature that contains information from the future, such as next week's stock-out flag or a price that was changed after the forecast was made, makes the test look very good and the real forecast fail
- **Noise.** A feature that carries no information about demand makes the model a little worse and much harder to explain. Weather at weekly product level is a common example
- **Cost.** Every feature has to be delivered every week, for every product, for years. A feature nobody maintains is a forecast that stops working
- **How to decide.** Start from what a planner would look at: history, promotions, calendar. Add one feature at a time and keep it only if the backtest error falls
- **How to check.** Scramble one feature and see how much worse the forecast gets. Scrambling the average of the last 26 weeks raises the error (WAPE) from 25 % to 68 %; scrambling the product identity raises it to 65 %. Most of the other 24 features change it by less than 1 point

→ A short list the business can explain and supply beats a long list the model hardly uses.

The Same Four Weeks Compared

Plush Bear, store, the same four weeks, all three approaches. Average miss: how far off per week, in units. Bias: forecast minus actual on average.



Approach	Average miss	Bias
Repeat last year	54 units	-51 units (-17 %)
Smooth recent demand	155 units	-155 units (-50 %)
Global model	56 units	-56 units (-18 %)

All three forecast too low this December. Repeating last year and the global model miss by about the same; the smoothed level misses by almost three times as much. Would you choose the same for the summer, or for the online channel?

→ One product and one window are a starting point, not a verdict.

Models Trained On Many Datasets

Foundation models: trained once on millions of time series, then used on your products without any training on them (zero-shot).

Model	Who	What is special
TimesFM	Google	One of the first; trained on a very large collection of time series
Chronos	Amazon	Treats a time series like a sentence of words
Moirai	Salesforce	Handles daily, weekly and monthly data and extra features in one model
TabPFN-TS	Prior Labs	A general model for tables, adapted to forecasting
TiRex	NXAI, Linz	Built on xLSTM, a neural network for sequences, from Sepp Hochreiter's group with JKU's LIT AI Lab; first in public zero-shot comparisons in 2025

- The extreme case of a global model: trained once on everything, no fitting on your data, and it returns a range, not only one number
- What to watch: most cannot use promotion or price features, slow movers are not their strength, and they must pass the same backtest as every other method

→ **Treat a new model as another candidate for the same test.**

Choosing A Method

Situation	Start with	Then try	Avoid
Smooth, seasonal, 2+ years of history	Seasonal naive as the benchmark; SES	Holt-Winters; the global model with promotion and calendar features	Choosing a method without a backtest
Smooth, 1 to 2 years of history	SES, or the average of the last 8 weeks (a moving average)	The global model, once the product has a year of history	Holt-Winters (too little history to learn the season)
Erratic	SES or a moving average	The global model; Holt-Winters with a damped trend	Seasonal naive alone (it copies last year's random ups and downs)
Intermittent	TSB or Croston SBA	The global model; judge the total over the lead time	Holt-Winters; choosing by a weekly percentage error (MAPE, chapter 3)
Lumpy	TSB or Croston SBA for the rate; plan the total over the lead time	The global model; compare by the stock outcome	Any choice based on weekly accuracy
Strong promotion or price effects	The global model with planned features	Holt-Winters on the sales without promotions, plus a separate promotion effect	Local methods alone
New product	Similar products, the category average, judgement	A global model built for it (ours needs a year of history)	Any method that needs the product's own history

→ **Start simple, keep the benchmark, and let the backtest decide.**

Hands-On 1: Choosing A Forecast

Open **Notebook 1**, sections 1 to 3. Copy to Drive, Run all, work in pairs.

- Section 1: read the channel picture and the four example products
- Section 2: the Christmas comparison for the Plush Bear. Then change the window to Summer, and the product to Garden Hose
- Try Plush Bear in store and online. Does your choice change with the channel?
- Section 3: write a choice, one supporting number and one limitation, and explain it to your partner
- Stuck? Runtime, Restart session and run all. Still stuck? Raise a hand, that is what I am here for

→ **Minimum: one comparison, and explain your choice to your partner.**

Evaluate

Metrics, honest tests, tuning, and
the model's reasons

03

The Size Of The Miss: MAE And WAPE

A teaching example with round numbers: four weeks, one forecast. $|x|$ means the size of x without its sign.

Week	Actual	Forecast	Miss = forecast - actual	Miss
1	100	90	-10	10
2	120	100	-20	20
3	80	70	-10	10
4	100	100	0	0
Total	400	360	-40	40

MAE, mean absolute error

- In words: add up the sizes of the misses and divide by the number of weeks
- $40 \div 4 = \mathbf{10 \text{ units}}$ a week

WAPE, weighted absolute percentage error

- In words: total size of the misses divided by total demand
- $40 \div 400 = \mathbf{10 \%}$. Comparable across products of different size

→ Take the size of each miss before averaging, so highs and lows do not cancel.

Accuracy And Direction Together

Two forecasts with the same total miss of 40 units on the same 400 units of demand.

	Forecast A	Forecast B
Weekly forecasts (actual 100, 120, 80, 100)	90, 100, 70, 100	110, 100, 90, 100
Total miss	40 units	40 units
WAPE: total miss ÷ total demand	10 %	10 %
Bias: (total forecast - total demand) ÷ total demand	-10 %: never too high	0 %: too high and too low cancel

- **Bias** in words: the average direction of the miss. Positive means the forecast runs too high, negative too low
- A stock rule fed by forecast A runs out; fed by forecast B it holds too much in some weeks and too little in others. Same accuracy number, different shelves

→ Same WAPE does not mean the same inventory consequences.

Accuracy On A Slow Mover

A slow mover, with round numbers: one sale in four weeks, and a forecast that always says zero.

	Week 1	Week 2	Week 3	Week 4
Actual units	0	0	0	4
Forecast units	0	0	0	0
Correct?	yes	yes	yes	no

MAPE, mean absolute percentage error

- In words: each week's $|\text{miss}|$ divided by that week's demand, averaged over the weeks
- Dividing by zero demand: **undefined** in three of four weeks

WAPE

- Total $|\text{miss}|$ 4 $\div$ total demand 4 = **100 %**: every unit of demand missed
- Three correct weeks and an empty shelf. For products with many zero weeks, judge the stock decision, not the share of correct weeks

→ Judge a forecast by the demand it covers, not by how many weeks it was right.

The Metrics In Words

Five numbers you will meet in every forecast report. $|x|$ is the size of x without its sign. Each metric is blind somewhere.

Metric	How it is computed	In words	Blind when	Use it for
MAE (mean absolute error)	average of $ \text{actual} - \text{forecast} $	the typical miss, in units	products of different size are compared	one product, reported to its planner
Bias	$(\text{total forecast} - \text{total actual}) \div \text{total actual}$	the average direction of the miss; positive means too high	on its own: it says nothing about accuracy	every table, next to the accuracy number
MAPE (mean absolute percentage error)	average of $ \text{actual} - \text{forecast} \div \text{actual}$	the typical miss as a percentage of that week's demand	any week with zero demand; weeks with low demand; it favours forecasts that are too low	a single fast mover, if at all
WAPE (weighted absolute percentage error)	$\text{sum of } \text{actual} - \text{forecast} \div \text{sum of actual}$	the total miss as a percentage of total demand	small products disappear behind big ones	a category or an assortment: the working default
MASE (mean absolute scaled error)	$\text{MAE} \div \text{MAE of the naive forecast (each week = the week before) on the training history}$	the miss compared with the past miss of simply repeating last week; below 1 beats it	a flat training history; fine with zero weeks	comparing methods across all kinds of products

→ **Never accept an accuracy number without the bias next to it.**

Backtest: Replaying The Past

A backtest replays the past: stand at a date, forecast the next weeks with what was known then, compare, move on.

Step	What we do	In our backtest
1. Stand at a past date	Pick a forecast date (the origin: the last week with known sales); train every method on the weeks up to it only	Origins every 4 weeks from 23 December 2024 to 24 November 2025
2. Forecast the next weeks	Forecast the 4 weeks after it (the test window) with every method, and score them	4 weeks ahead, all 80 series, flagged weeks left out
3. Move forward and repeat	Collect the misses over the whole year	13 test dates, 4,160 forecasts per method, 4,158 scored

- Never assign weeks at random to training and test: the model would have seen the weeks right before and after, and the test is too easy
- One test window is one sample: the four weeks before Christmas rank the methods differently from four weeks in June

→ **Ask: which dates, how far ahead, and what was known at the time?**

Backtest Schemes

Scheme	How	Good for	Weakness
Fixed split	One date; train before, test after	A first look	One test window: Christmas ranks methods differently from June
Rolling origin	Many dates; at each one, train on everything before and test on the weeks after	The standard; it works the way forecasts are used in practice	More work: every method is retrained at every date
Sliding window	Like rolling origin, but train only on a fixed number of recent weeks, for example the last 104	A business that changed	Forgets the history the season needs
Random split	Weeks assigned at random to training or test	Never for time series	The model has seen the weeks right before and after the test week, so the test is too easy

- Ours is a rolling origin: 13 test dates, 4 weeks ahead, every method retrained at every date, flagged weeks left out of the scoring

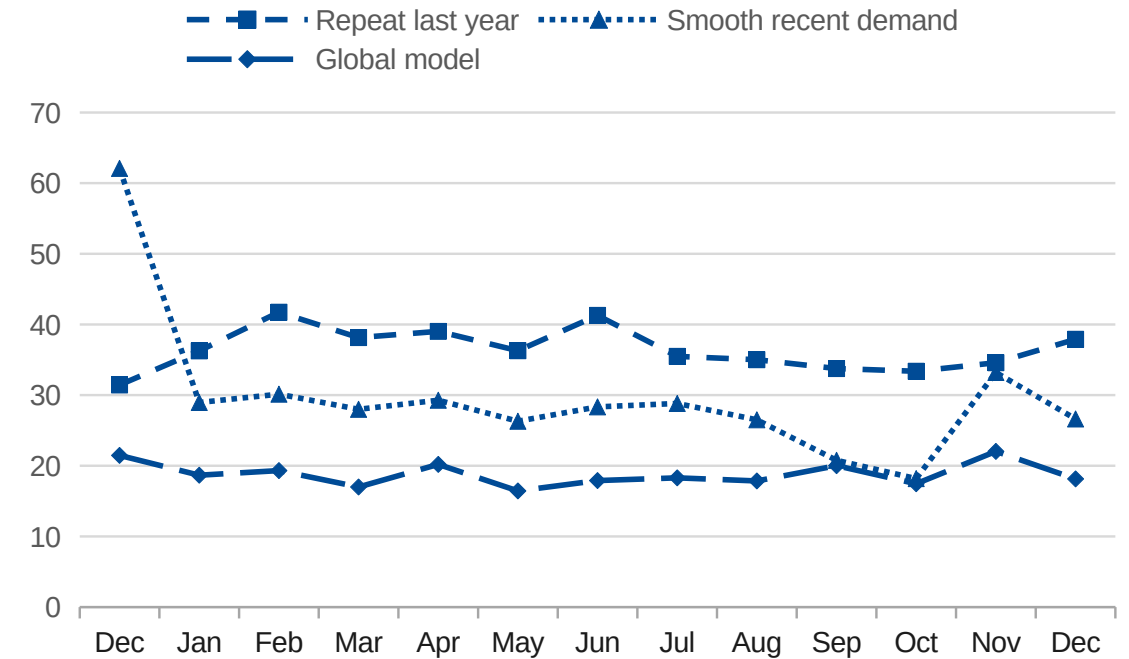
→ **When someone shows you a comparison of methods, ask two things: how many test dates, and which weeks of the year.**

Fixed Split And Rolling Origin

Smooth series: the same methods judged on one test window (the four weeks from 1 December 2025) and on all 13 windows.

Method	One window	Rank	All windows	Rank
Global model	18.1 %	1	18.8 %	1
Croston SBA	34.1 %	6	29.2 %	2
Smooth recent demand	26.6 %	2	29.4 %	3
TSB	33.2 %	5	30.3 %	4
Moving average (8 weeks)	31.5 %	3	31.6 %	5
Holt-Winters	33.0 %	4	32.4 %	6
Repeat last year	37.9 %	7	36.6 %	7

- Croston SBA is rank 6 on the four weeks before Christmas and rank 2 over the whole year. The global model is first in both tests



WAPE per test window in percent, smooth series, labelled by the month of the first test week (30 Dec 2024 to 1 Dec 2025).

→ One test window is one sample: only the first and the last place stay the same when the test dates change.

Backtest Results Across The Year

WAPE and bias over all 80 series and 13 test dates, then WAPE by kind of demand. Bold marks the lowest WAPE in each column.

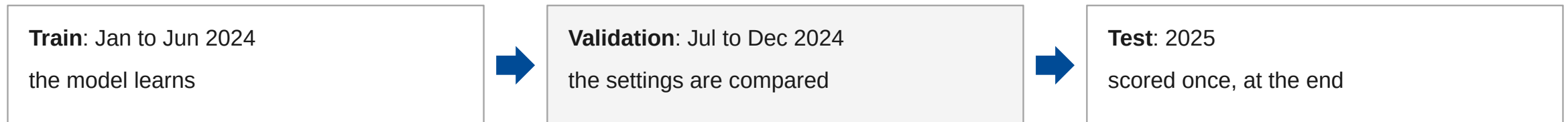
Method	WAPE, all	Bias, all	WAPE Smooth	WAPE Erratic	WAPE Intermittent	WAPE Lumpy
Repeat last year	43.8 %	-2.7 %	37 %	95 %	145 %	194 %
Moving average (8 weeks)	37.5 %	+0.2 %	32 %	79 %	135 %	162 %
Smooth recent demand	34.9 %	-0.9 %	29 %	74 %	129 %	160 %
Holt-Winters	39.0 %	-1.0 %	32 %	86 %	143 %	189 %
Croston SBA	34.9 %	-4.9 %	29 %	76 %	128 %	158 %
TSB	36.1 %	+0.0 %	30 %	77 %	132 %	162 %
Global model	25.1 %	-3.1 %	19 %	71 %	121 %	137 %

- The global model has the lowest WAPE in every column (19 % on smooth series against 37 % for repeating last year), but runs low wherever sales are irregular (-16 % erratic, -15 % intermittent, -35 % lumpy)
 - Among the local methods, simple smoothing (Smooth recent demand 34.9 %, Croston SBA 34.9 %) beats the more complex Holt-Winters (39.0 %)
- **An assortment average gives large sellers more influence; read the pattern columns and the bias too.**

Tuning The Model's Settings

Parameters are learned from the data. Hyperparameters are the settings chosen before training; tuning searches for good ones.

Model	Learned from the data (parameters)	Chosen before training (hyperparameters)
Exponential smoothing	the level; the smoothing weight α is fitted as well	trend or not, damping or not, season length of 52 weeks
Global model (LightGBM)	every decision rule (split) and end value (leaf) of every tree	number of trees; learning rate (how much each new tree corrects the ones before); tree size (leaves per tree, rows per leaf); share of rows and of features each tree sees; Tweedie power (how it treats weeks with zero and small sales)

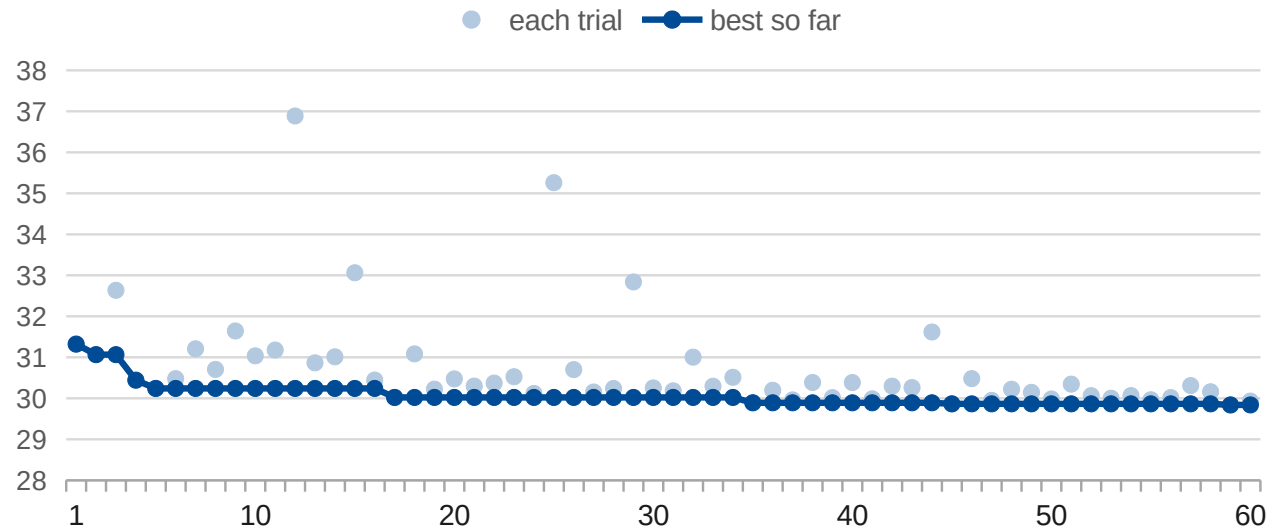


- **The goal:** the settings with the smallest error on weeks the model did not learn from
- **The search:** a grid tries every combination, a random search tries a sample, and Optuna, a free Python library, learns from each finished trial (one training run with one setting) where to look next

→ **Tune on the validation weeks; the test year is used once, at the end.**

Tuning With Optuna

60 trials: each trains the global model with one set of settings on Jan to Jun 2024 and scores it on Jul to Dec 2024 (WAPE, 1 to 4 weeks ahead).



Setting	Our setting	Tuned
learning rate	0.030	0.022
number of trees	600	350
leaves per tree	31	27
minimum rows per leaf	20	15
share of rows per tree	0.80	0.92
share of features per tree	0.80	0.43
Tweedie power	1.30	1.65
WAPE, validation weeks	31.3 %	29.8 %
WAPE, 2025, one training	25.5 %	25.6 %

Vibe-code it yourself: describe the task in plain words and let Gemini in Colab write the code

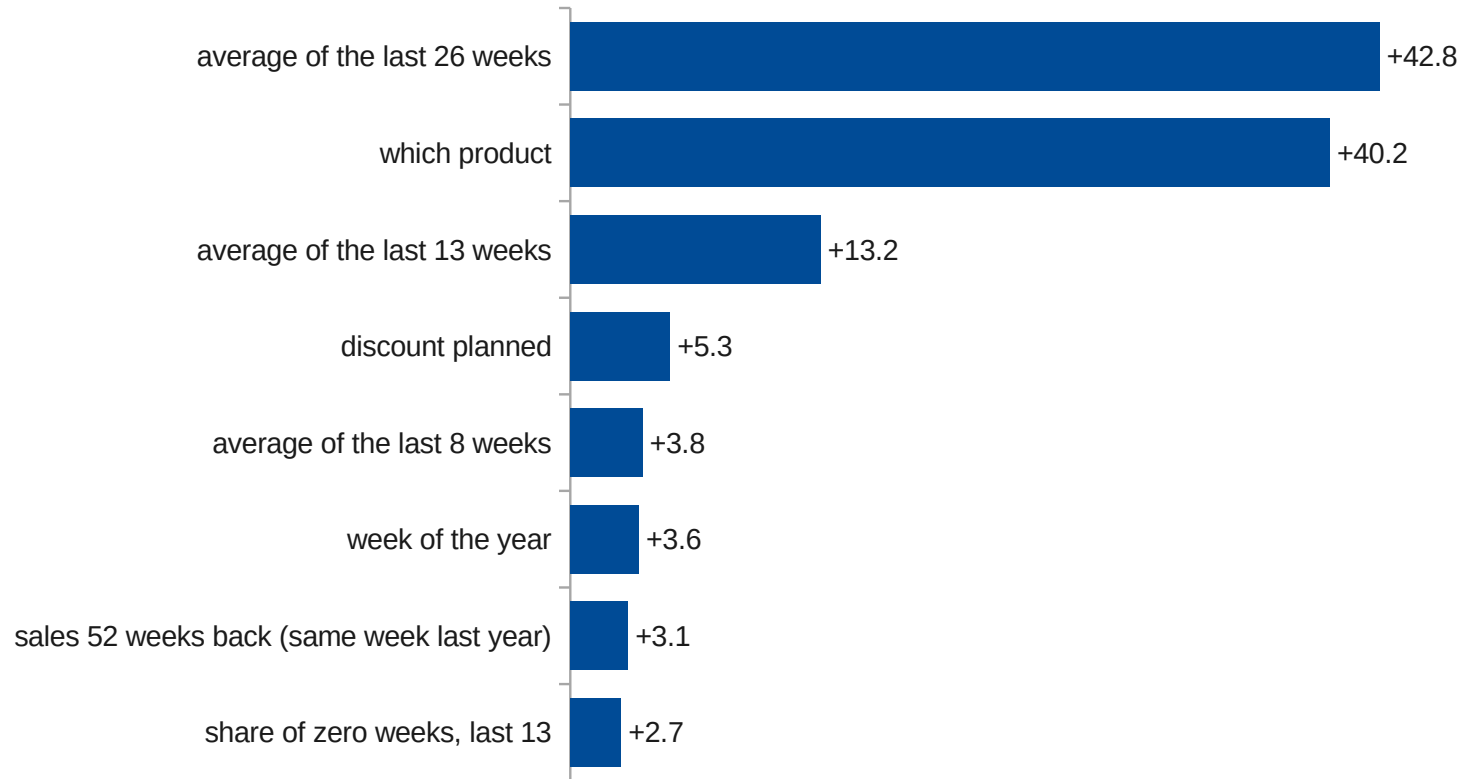
Write Python code that tunes a LightGBM forecasting model with Optuna. Train on January to June 2024 (2023 only feeds the lag features), score every trial by WAPE on July to December 2024, start from learning rate 0.03, 600 trees, 31 leaves, 20 rows per leaf, 0.8 row and feature shares and Tweedie power 1.3, run 60 trials, never use data from 2025, and print the best settings next to these.

→ **Tuning lowered the validation WAPE by 1.5 points and did not help on the test year: our settings were already good.**

What The Model Relies On

Permutation importance: scramble one feature in the 2025 data and see how much worse the forecast gets.

Without scrambling: WAPE 25.1 %.



The 8 largest of 26 features. Increase in WAPE, percentage points, average of 3 scrambles.

→ **These are the model's habits, not the causes of demand.**

- The model relies most on the product's **recent average sales** and on **which product** it is: scramble either and the error rises from 25 % to 65 to 68 %, more than two and a half times
 - **Discount** and **promotion** rank lower because they are on in about one week in 12; when they are on, they change the number a lot
 - Of 26 features, most cost under one point. They could be removed, and the model would be easier to explain
- A short, explainable list of features is a strength, not a weakness

One Forecast Explained

Plush Bear, store, week of 1 December 2025, one week ahead, from the model trained once on data up to 2024: forecast 274, actual 296.

Feature	Its value	Factor
which product	Plush Bear	x 4.16
average of the last 26 weeks	140	x 3.99
average of the last 13 weeks	141	x 1.34
week of the year	49	x 1.22
sales 52 weeks back (same week last year)	228	x 1.12
average of the last 8 weeks	146	x 1.09
all other features together		x 1.34

Start from the model's **reference level of 6.17 units** (its starting value: a typical week across all 80 series, most of them small sellers) and multiply by every factor: 274 units. No promotion this week: the promotion and discount factors are close to 1 and sit in the last row.

In a meeting: "We forecast 274. The bear's recent weeks and the December week explain most of the number; actual demand was 296, so our forecast was 22 units too low."

→ A factor tells you how the model built the number, not why customers bought.

From Forecast To Stock

The protection period, safety stock,
one order, and what a service
target buys

04

From A Weekly Forecast To Stock

Term	Plain meaning	Our example
Lead time	The wait between placing an order and receiving it	4 weeks
Review period	How often we look at the stock and place an order	every week
Protection period	Lead time plus review period: the weeks the stock on hand and on order must cover	5 weeks

- Whatever we order today arrives in four weeks and has to last until the next delivery, one week later. So today's decision must cover five weeks of demand
- The next five slides build one order from that: add the forecasts, add a cushion, see where its size comes from, subtract what is already there

→ **First decide which weeks your stock must cover.**

Forecast Over The Protection Period

Round teaching numbers: five weekly forecasts for the five weeks the stock must cover.

Week ahead	1	2	3	4	5	Sum
Forecast, units	180	190	210	200	220	1,000

Forecast over the protection period: 1,000 units.

- If the forecast were perfect, 1,000 units on hand and on order would be exactly enough. It is not perfect, so we add a cushion next

→ **Use the sum over the protection period, not one week's value.**

Safety Stock

The cushion is called safety stock. It is extra stock, on top of the forecast.

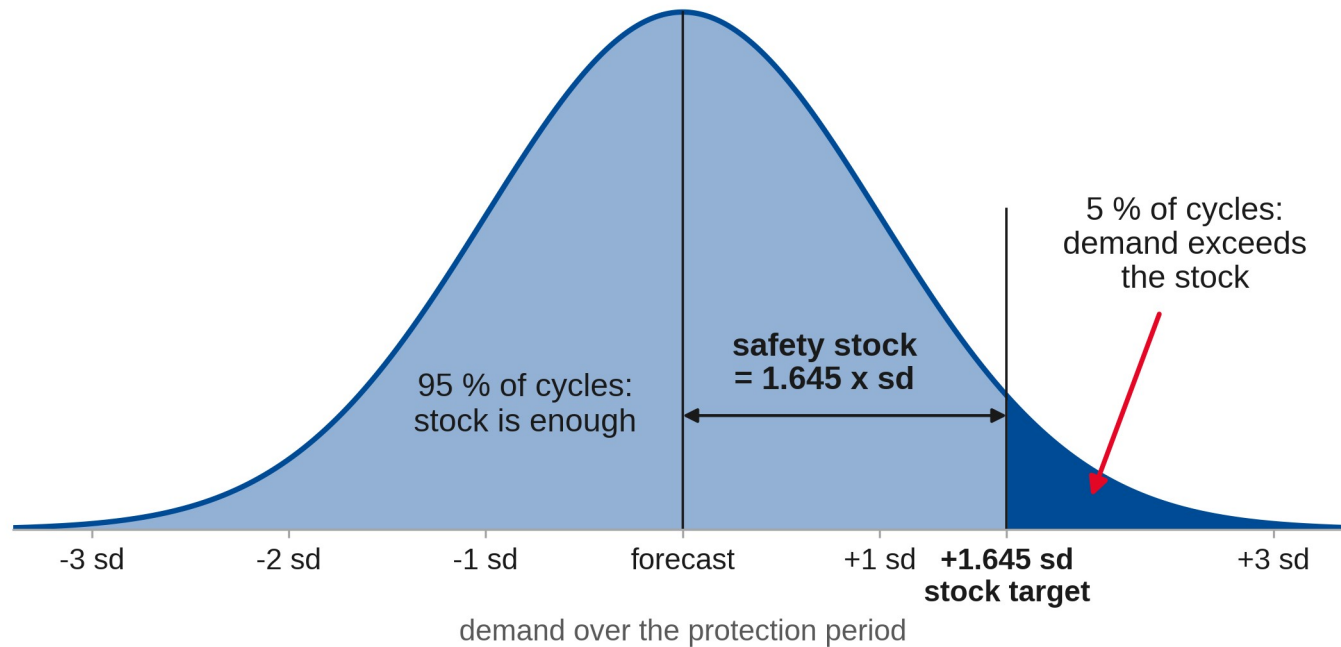
Quantity	Meaning	Value
Error spread (standard deviation, sd)	How widely the weekly misses scatter around their average, from the backtest; for a bell curve about 1.25 x the average miss (MAE)	40 units
Cycle service target	The share of order cycles (here one week, from one delivery to the next) that we plan to get through without a stock-out	95 %
Protection period	The weeks the cushion has to cover	5 weeks
Safety stock	z for the target x error spread x square root of the weeks: $1.645 \times 40 \times \sqrt{5} = 147.1$, rounded up to 148	148 units

- Why the square root: over five weeks the misses partly cancel out, so the spread grows by about 2.2 times, not 5 times. That assumes the weekly misses are independent
- Where does 1.645 come from? From the bell curve, next slide

→ **Safety stock depends on how wrong the forecast tends to be, and on your service target.**

The Safety Factor z

If the misses follow a bell curve, demand stays below the forecast plus z standard deviations (sd) in a fixed share of cycles. For a 95 percent target, $z = 1.645$.



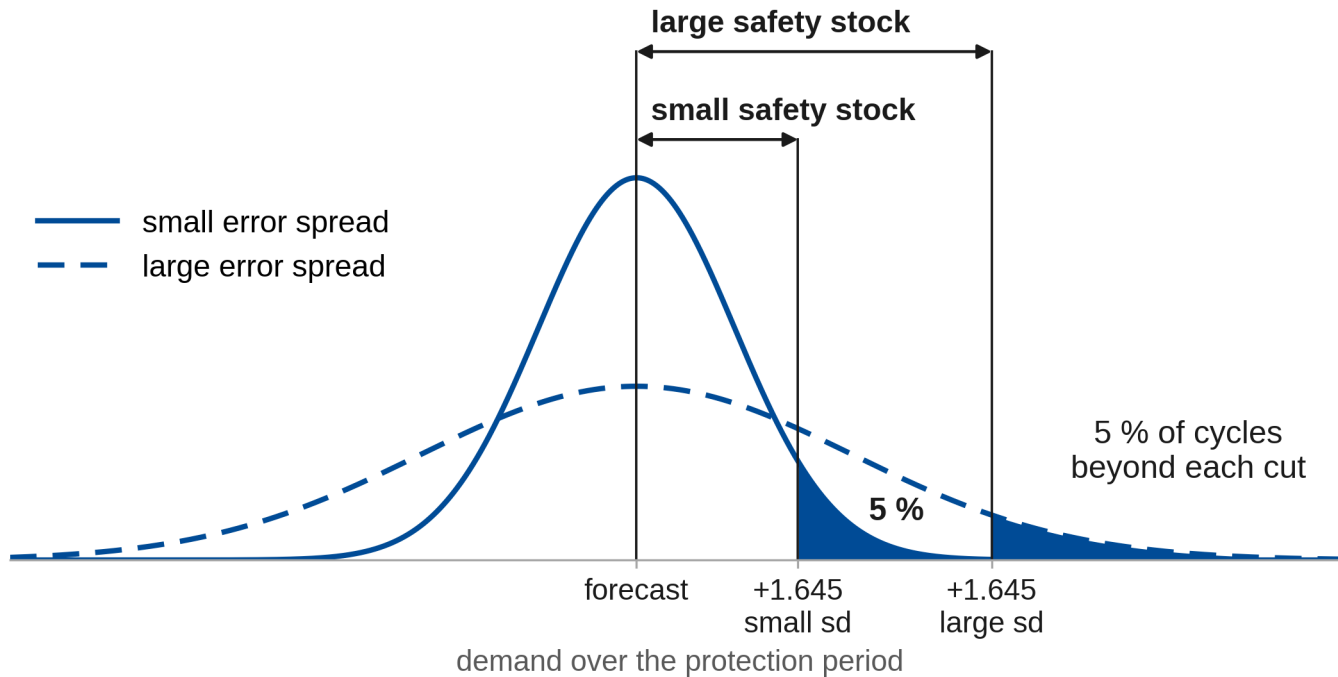
Target	z	Safety stock, our example
50 %	0.00	0 units
84 %	1.00	90 units
90 %	1.28	115 units
95 %	1.645	148 units
99 %	2.33	209 units

- Spread over the 5 weeks: $40 \times \sqrt{5} = 89.4$ units. Safety stock = $z \times 89.4$, rounded up to whole units
- From 95 to 99 percent the cushion grows by 41 percent for four points of service: the curve is thin at the edge

→ **z belongs to the bell curve; the spread is yours, from the backtest; the target is the business's choice.**

One z For Every Product

Every bell curve has the same shape. Measured in standard deviations (sd), 95 percent of cycles always stay below +1.645 sd.



Store, 95 % target	z	Spread over L + 1 weeks	Safety stock
Plush Bear	1.645	88.8 units	147 units
Puzzle 1000	1.645	67.3 units	111 units
Vacuum Filter	1.645	2.45 units	5 units

- The same z for every product: the cut sits 1.645 spreads above the forecast, and 5 percent of cycles lie beyond it
- What differs is the spread in units, so the safety stock differs: 147 units for the Bear, 5 for the Filter
- This holds if the misses follow a bell curve; the replay later in this chapter checks it

→ z counts standard deviations, not units: one table serves every product.

Order Quantity

The stock target (order-up-to level) is what we want on hand and on order together. We order only the difference.

Quantity	Units
Stock target = forecast over the protection period + safety stock	$1,000 + 148 = 1,148$
Already on the shelf + already on order	$300 + 500 = 800$
Order now = stock target - what is already there	$1,148 - 800 = 348 \text{ units}$

- Ordering the whole target would count the 800 units twice. Every week the same rule runs again with the new forecast and the new stock on the shelf and on order
- Assumptions worth saying aloud: fixed lead time, deliveries as scheduled, no backorders (a customer who finds the shelf empty does not wait: the sale is lost), and misses that follow a bell curve

→ **Count what is already on the shelf and already on order.**

Safety Stock For Three Products

Orders every week; protection period = lead time L plus one review week; safety stock = $z \times \text{weekly spread (sd)} \times \sqrt{L + 1}$, rounded up.

Product	Lead time	Units / week, Jul to Dec 2025	Weekly spread (sd)	Spread over L + 1 weeks	Safety stock at 90 %	Safety stock at 95 %	Safety stock at 99 %	Safety stock in weeks of demand, 95 %
Plush Bear, store	4 weeks	165.8	39.7	88.8	114	147	207	0.9
Puzzle 1000, store	5 weeks	33.9	27.5	67.3	87	111	157	3.3
Vacuum Filter, store	6 weeks	0.58	0.92	2.45	4	5	6	8.7

Weekly spread measured on January to June 2025, units per week over the 26 replay weeks.

- The Puzzle sells 20 percent of the Bear's volume but needs 76 percent of its safety stock: its weekly spread is almost as large as its sales
- For the Filter the safety stock is 8.7 weeks of demand. The z factors assume that the misses follow a bell curve. For a product that sells single units, this assumption is poor. The replay on the next slides is the better check
- $z = 1.28, 1.645$ and 2.33 for 90, 95 and 99 percent. Each step up the curve buys less service for more stock

→ **Every unit of error spread you remove saves $z \times$ the square root of the protection period in safety stock.**

Service Target Versus Fill Rate

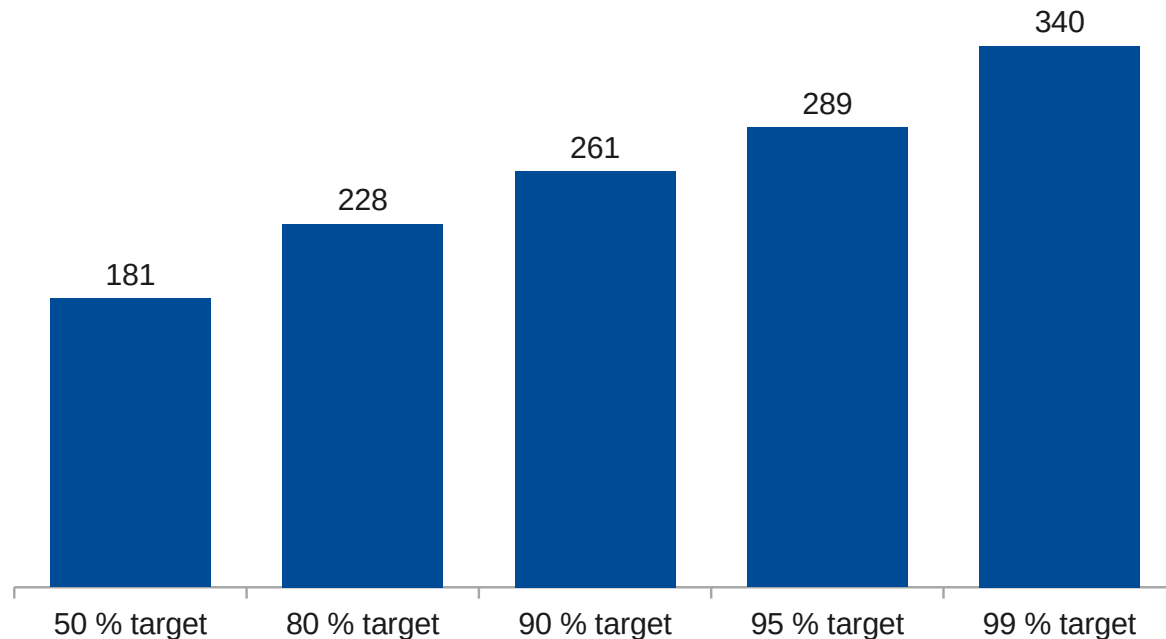
Measure	The question it answers	Target or result?
Cycle service target	How likely do we want each order cycle (one week, from one order to the next) to end without a stock-out?	A target we choose
Fill rate	What share of the units customers asked for did we actually serve?	A result we measure
Average stock on the shelf	How much stock did the rule hold, on average?	A result we measure
Weeks with a stock-out	How often did demand exceed the stock available?	A result we measure

- How we measure the fill rate: replay the ordering rule week by week over 26 weeks; each week, served = the smaller of demand and stock on hand; fill rate = total units served ÷ total units demanded over the 26 weeks

→ A 95 percent cycle service target does not mean 95 percent of units served.

Effect Of A Higher Service Target

Plush Bear, store: the ordering rule replayed week by week, 2025-06-30 to 2025-12-22, at five service targets.



Average units on the shelf.

Target	Fill rate	Stock-out weeks	Units not served
50 %	98.0 %	7	85
80 %	100.0 %	0	0
90 %	100.0 %	0	0
95 %	100.0 %	0	0
99 %	100.0 %	0	0

From 50 to 80 percent: 2.0 points more demand served for 47 more units on the shelf. From 95 to 99: 0.0 points for 51 units. Above the 80 percent target, every step adds stock and no service in these 26 weeks.

→ Compare added stock with added service before raising the target.

Same Target, Different Outcomes

Three products, the same 95 percent target, the same rule, replayed over the same 26 weeks.

Product	Lead time	Error spread (sd)	Fill rate	Stock-out weeks	Average stock, weeks of demand
Plush Bear, store	4 weeks	39.7 units	100.0 %	0	1.7
Puzzle 1000, store	5 weeks	27.5 units	87.2 %	6	1.9
Vacuum Filter, store	6 weeks	0.92 units	93.3 %	1	7.5

Average stock on the shelf ÷ average weekly demand, 26 replay weeks.

- **Plush Bear:** no stock-out, 1.7 weeks of demand on the shelf. Its forecast ran about 9 percent high, which added to the cushion
- **Puzzle:** a 95 percent target and only 87 percent of the units served. Its forecast ran about a third too low in the test months, and the safety stock covers only the spread of the misses, not a bias
- **Vacuum Filter:** 7.5 weeks of demand on the shelf, in about 4 units. Normal for a slow mover with a 6-week lead time; check the unit cost before calling it too much

→ **Inspect the product and its costs before calling stock excessive.**

Hands-On 2: Choosing A Service Target

Open **Notebook 2**, sections 1 to 4. Copy to Drive, Run all, work in pairs.

- Section 1: the worked order, the same numbers as on the slides
- Section 2: move the target slider for the Plush Bear: 50, 80, 95, 99 percent. What does each step buy?
- Section 3: compare the three products at 95 percent. Then choose the Puzzle or the Filter in the section 2 slider: does the same target still make sense?
- Section 4: write your decision in four lines: the target you chose, the stock and fill rate you observed, what you accept, what you would check first
- Stuck? Runtime, Restart session and run all. Still stuck? Raise a hand, that is what I am here for

→ **Minimum: one product, two service targets, one defended choice.**

Holding Cost Of Each Forecast

Plush Bear, store, 95 percent target, the same rule: only the forecast changes, and each method brings its own error spread.

Forecast	Error spread (sd)	Average stock	Fill rate	Holding cost per year
Repeat last year	77.7 units	351 units	100.0 %	EUR 971
Smooth recent demand	59.6 units	208 units	88.0 %	EUR 574
Global model	39.7 units	289 units	100.0 %	EUR 798

Unit cost EUR 13.83, holding rate 20 percent of the unit cost per year. Shortage costs not included.

- A better forecast has a smaller error spread, so it needs a smaller cushion: the global model serves all demand with 82 percent of the stock that Repeat last year needs
- Smooth recent demand holds the least stock because its flat forecast misses the Christmas rise: it runs out in all 4 December weeks and serves 88 percent of demand. It is cheaper to hold, but it is a different policy, not a better one

→ **Read stock, service and cost together.**

Key Takeaways

- **Start with the pattern.** How often a product sells and how regular the amounts are tell you what error to expect and which methods to try, before any model
- **A benchmark in every table.** Keep repeating last year in every comparison: a method that cannot beat it is not worth its complexity. Over our test year, every method beat it
- **One model for all products wins on every kind of demand** (smooth series: WAPE 19 % against 37 % for repeating last year), but runs low wherever sales are irregular. It learns from all products at once, including the planned promotions
- **Every accuracy number is blind somewhere.** MAPE is undefined when demand is zero and rewards low forecasts; WAPE hides small products. Never read accuracy without bias
- **Ask how it was tested.** How many test dates, which weeks of the year, and whether future information leaked in (leakage)
- **Features carry the knowledge.** A series becomes lags, averages, calendar and plan; tune the settings on validation weeks and keep the test year for the end
- **A model has reasons.** Ask for them as factors; "the model expects this December week to add about 22 percent" is a sentence you can check
- **Judge a forecast by the stock it needs.** Safety stock is z times the error spread over the protection period, so a better forecast means the same service with less stock

- **What will we order, and what evidence makes that a reasonable choice?**

Further Reading And Tools

The notebooks and the data stay at github.com/FH-Prevail/jku_course. Most of the reading is free.

If you want to	Start here
Open the notebooks again	Notebook 1 in Colab and Notebook 2 in Colab ; in Colab, File, Save a copy in Drive
Understand the methods properly	Hyndman, Athanasopoulos: Forecasting: Principles and Practice, 3rd ed., otexts.com/fpp3 , free online
Read about slow movers	Syntetos, Boylan (2005), Int. J. Forecasting 21; Teunter, Syntetos, Babai (2011), EJOR 214; Mirshahi et al. (2026), Two-Step Framework for Intermittent Demand Forecasting, Procedia CS 278
See a global model in a real competition	Makridakis, Spiliotis, Assimakopoulos (2022): The M5 competition, Int. J. Forecasting 38
Try a foundation model	github.com/NX-AI/tirex ; my comparison: Mirshahi et al. (2026), Forecasting Without Fine-Tuning, Procedia CS 277
Read about promotions in retail	Darbanian, Brandtner, Falatouri, Nasser, Mirshahi (2025): Timing Matters, J. Retailing and Consumer Services 85
Go deeper on inventory	Silver, Pyke, Thomas: Inventory and Production Management in Supply Chains, chapters on safety stock and on the newsvendor problem (one order for one selling period)
Ask me anything	Sina.Mirshahi@fh-steyr.at

Thank you



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Notebooks and data:
github.com/FH-Prevail/jku_course →



Reference: One Christmas Order

Newsvendor: one order for one selling period. Puzzle 1000, store, the week of 15 December 2025.

Cost recovered on a leftover	Cost of a leftover	Critical ratio	Order
0 % of the purchase cost	EUR 38.86	0.37	80 units
50 % of the purchase cost	EUR 19.43	0.54	92 units
95 % of the purchase cost	EUR 1.94	0.92	128 units

- Forecast **88 units**, error spread 27. A unit short costs the lost margin, EUR 23.23
 - Critical ratio = cost short ÷ (cost short + cost of a leftover). The order is the demand level that is not exceeded with that probability: a **quantile**
 - Actual demand that week: 41 units
- If you do not know the costs, you cannot recommend a quantity

→ The forecast stays fixed; the cost of leftovers changes the order.

Reference: The Limits Of Our Replay

- The sales are artificial; the four flagged stock-out weeks were estimated for training and left out of the scoring
- Orders are whole units; supplier lead times are fixed; deliveries arrive as scheduled; there are no backorders: demand we cannot serve is lost, not delivered later
- The safety stock rule assumes stable, unbiased, independent misses that follow a bell curve. For sparse demand that is a poor assumption
- The error spread was measured on 25 weeks and the rule replayed on 26 weeks; costs, bias and changing conditions still need review
- The global model needs a year of history per product; it cannot forecast a new product that has no sales history yet

→ **A reproducible classroom result is evidence, not a guarantee for a retailer.**